

PROJECT TITLE



by

Enter Student Name 1

Enter Student Name 2

Supervisor

Enter Supervisor Name

Co-supervisor(s)

Enter Co-supervisor Name 1

Enter Co-supervisor Name 2

Department of Mechanical Engineering
Institute of Space Technology, Islamabad
Month 20XX

TITLE

A thesis submitted to the
Institute of Space Technology
in partial fulfillment of the requirements
for the degree of Mechanical
Engineering

by

Enter Student Name 1 – Registration Number

Enter Student Name 2 – Registration Number

Supervisor

Enter Supervisor Name

Co-supervisor

Enter Co-supervisor Name 1

Enter Co-supervisor Name 2

Department of Mechanical Engineering
Institute of Space Technology, Islamabad
Month 20XX

Institute of Space Technology
Department of Mechanical Engineering



TITLE

by

Enter Student Name 1 – Registration Number

Enter Student Name 2 – Registration Number

APPROVAL BY BOARD OF EXAMINERS

Supervisor:

Enter Supervisor Name

Co-Supervisor: Enter Name

Co-Supervisor: Enter Name

AUTHORS DECLARATION

We declare complete responsibility and accountability for the work conducted under the project titled.

“Title”

The work completed as part of this project is the outcome of our independent study and has been carried out under the guidance and supervision of the supervisors and co-supervisors of the project. Any support or contributions from other people or sources have been acknowledged and properly cited. The ideas, approaches, algorithms, and models associated with the project are rightly referred to and cited from existing literature and all relevant sources. A comprehensive bibliography has been provided, including all the details regarding the sources consulted. The results and conclusion put forth in this thesis are dependent on thorough analysis and careful interpretation of the collected data. The assumptions associated with the analyses and the limitations catered during the research work are properly acknowledged and discussed.

No other academic degree or certification has received this thesis, in whole or in part. It is a unique piece of work created only to satisfy the requirements for the degree of Bachelor of Science at the Institute of Space Technology.

I am aware that any inaccuracies, misrepresentations, or instances of plagiarism in this declaration could have serious repercussions, such as the invalidation of this thesis and disciplinary action in accordance with the policies approved by the Institute of Space Technology.

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I acknowledge that the submitted thesis is in its final form and should be scrutinized for plagiarism as per IST policy.

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Dated_____

CERTIFICATE

This is to certify that I have carefully read, assessed, and approved the contents of this thesis and that it satisfies my standards for academic diligence, research technique, and adherence to the standards set by the Institute of Space Technology. All the information, findings, and materials reported in this manuscript have been personally reviewed by me, and I am confident in stating that they are accurate and genuine. The thesis effectually presents the research gap, research objectives, methodology, and results. The content has not been plagiarized, and any used material has been properly cited. Based on my evaluation, I endorse the acceptance of this thesis for the award of Bachelor of Science in Mechanical Engineering. I strongly affirm that the work matches the standard set by the Examination Committee.

(Enter Supervisor Name)

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DEDICATION

- Dedicate your thesis to someone who significantly impacted your research or supported your journey. (Who and why)
- Keep it concise. Express your gratitude for their specific contributions. (Be Specific and Sincere)
- Optional: Include a meaningful quote or short anecdote to personalize the dedication.

ACKNOWLEDGEMENT

- Acknowledge individuals and institutions that provided significant support during your research. (Who and What)
- Express your gratitude for their contributions (funding, guidance, resources). (Be Specific and Thankful)
- Maintain a professional tone while allowing some personality to shine through.

ABSTRACT

The **abstract** summarizes the entire thesis in about **200-250 words**. It should:

- **State the research problem:** Begin with a one-sentence description of the issue or gap the thesis addresses.
- **Outline the methodology:** Mention the key methods (e.g., experimental, numerical simulations, or theoretical modeling).
- **Summarize key findings:** Highlight the most important results.
- **Conclude with implications:** Discuss the significance of the findings briefly.

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LIST OF ABBREVIATIONS

CFD	Computational Fluid Dynamics
ANN	Artificial Neural Network
HAWT	Horizontal Axis Wind Turbine
VAWT	Vertical Axis Wind Turbine
MOPSO	Multi-Objective Particle Swarm Optimization
CAD	Computer-Aided Design
GA	Genetic Algorithm
RANS	Reynolds-Averaged Navier-Stokes
SIMPLE	Semi-Implicit Method for Pressure-Linked Equations

1. INTRODUCTION

1.1.Motivation

- Begin with a compelling fact, statistic, or statement that highlights the significance of the research area. This opening should immediately capture the reader's interest.
- Explain how the research topic relates to pressing challenges or emerging trends in mechanical engineering. Establish its relevance within the field.
- If applicable, mention any personal or academic drive for selecting the research topic, ensuring a professional tone. This adds a personal dimension without detracting from the technical focus.
- Ensure the section is concise and directly addresses the *why* behind the research. Avoid detailed discussions of methodology, objectives, or findings, as these belong in later sections.
- Write in a formal, objective tone, avoiding overly emotional or informal language. The motivation should set the stage for the research's importance while engaging the reader effectively.

1.2.Background

Provide the broader context of the research area. Why is the research topic important in mechanical engineering? Discuss current trends or challenges that make your study relevant.

1.3.Objectives

List the specific objectives of the study. These should directly address the research problem.

1.4.Scope

Explain the boundaries and focus of the study, including what is included and what is not. This helps define the limits of the research.

1.5. Environment and Sustainability

- Highlight specific environmental issues directly related to the research topic, such as pollution, high energy consumption, resource inefficiency, or waste generation.
- Explain how the research influences environmental factors, either positively (e.g., reduced emissions) or negatively (e.g., resource use during implementation).
- Describe how the study supports sustainability through actions like improving energy efficiency, conserving resources, or reducing waste and emissions.
- Assess how the outcomes of the research might influence the environment in the short and long term.

1.5.1. Relevance to Sustainable Development Goals (SDGs)

- Clearly mention the specific SDGs the research addresses and briefly describe their significance.
- Link the technical findings of the research to global SDG targets.
Example: "The design improvements align with SDG Target 7.3, which focuses on doubling the global rate of improvement in energy efficiency, and indirectly support SDG Target 9.4, promoting sustainable industrialization."
- Explain how the research fosters global efforts to achieve long-term sustainability beyond specific SDGs.
- Highlight how the findings can influence industry practices, policy-making, or societal habits.

1.6. Thesis Outline

- Provide a summary of the chapters that follow the **Introduction**, giving the reader a clear roadmap of the thesis structure.
- Keep the description concise, focusing on the purpose and content of each chapter rather than specific details.

- Ensure the outline flows logically, reflecting the progression of the research from context and methods to results and conclusions.
- Use formal and professional language, avoiding unnecessary repetition of chapter titles or overly technical terms.

2. LITERATURE SURVEY

2.1. Historical Background

- Summarize key research in the area, discussing both theoretical and experimental work.
- Highlight major contributions by other researchers.
- Critically examine the methodologies, results, and conclusions of previous studies. Discuss strengths and weaknesses

2.2. Research Gaps

- Identify gaps in the existing literature where knowledge is incomplete or conflicting. This section directly links to the problem statement by showing the need for further investigation.
- Explain your research's potential impact and contribution to filling this gap.
- Briefly mention other potential ways to address the gap if applicable. (Other Perspectives - Optional)

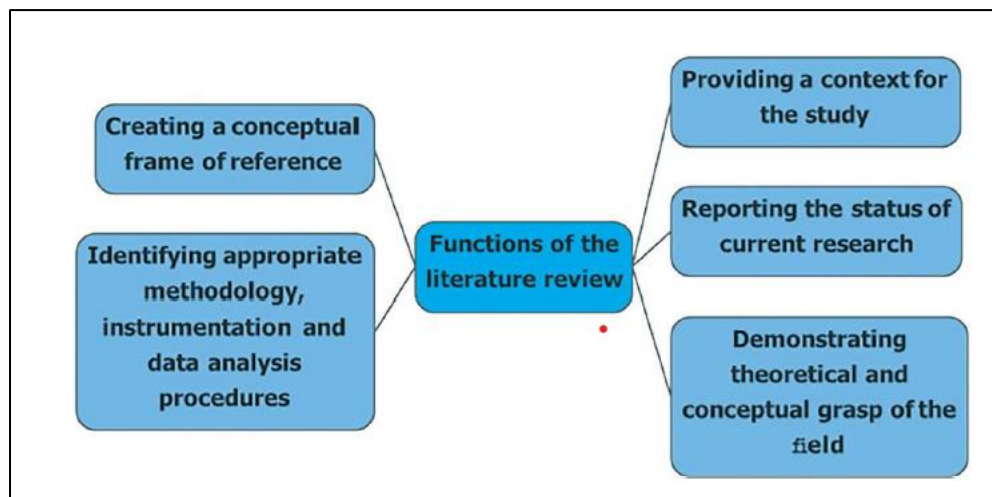


Figure 2-1: Functions of the Literature Review

2.3.Problem Statement

Clearly define the specific problem or research question the thesis aims to address. It should be specific and directly related to the research.

Sample Referencing Style:

In 2014, Daryoush Allaei and Yiannis Andreopoulos presented a new design for harnessing wind energy: an INVELOX wind turbine. This turbine provided solutions to all the major problems related to conventional wind turbines, such as reliability, intermittency issues, and wildlife hazards. The system's omnidirectionality resulted in a significant increase in power output [1].

3. METHODOLOGY

- **Research design:** Explain the overall approach, whether experimental, computational, or theoretical. Justify the selection of methods based on the problem at hand.
- **Data Collection:** Specify how data was collected through direct measurement (for experiments) or output from simulations. Mention any important parameters.
- **Experimental Setup or Simulation Tools:** Describe the tools, equipment, or software used. If conducting simulations, mention the specific software (e.g., ANSYS Fluent) and provide details on the setup (e.g., boundary conditions, mesh generation).
- **Address ethical considerations:** Outline the steps taken to ensure ethical research practice (informed consent, data privacy).
- **Mention methodology limitations:** Discuss any limitations or potential biases in your chosen methods.
- **Present results visually and descriptively:** Present your important findings using tables, figures, and clear explanations.

4. RESULTS AND DISCUSSION

- **Data Presentation:** Present your findings in an organized manner using graphs, tables, or charts. Ensure each figure/table is labeled clearly and referenced in the text.
- **Discuss key findings and observations:** Highlight the most important results and any unexpected observations.
- **Compare findings with existing literature:** Relate your results to previous research findings, highlighting any agreements or discrepancies.
- **Explain the limitations of your study:** Discuss any limitations of your experimental design or data analysis that could impact the conclusions.

Table 4.1: Caption

Sr. No.	Input	Output
1		
2		

5. CONCLUSION AND FUTURE WORK

- **Restate your research objectives/questions:** Briefly remind the reader of your research goals and the key questions you addressed.
- **Summarize key findings:** Highlight the most significant findings and outcomes of your research in a concise manner.
- **Explain the implications of your findings:** Discuss the broader implications and significance of your research for your field or society.
- **Connect findings to research gap:** Explain how your findings address the research gap you identified earlier in your thesis.
- **Mention limitations and future research directions:** Acknowledge any limitations of your study and suggest potential areas for further investigation.
- **Conclude with a strong statement:** Restate the overall contribution of your research and its potential impact on the field.

REFERENCES

1. General Formatting Rules

- The reference list is placed at the end of the paper. (Use ENDNOTE for adding references)
- References are numbered in the order they are cited in the text.
- Each reference number is enclosed in square brackets.
- Use the author's initials followed by their surname (e.g., J. Smith).
- Italicize the book, journal, or conference title.
- Use abbreviations for common words in journal or conference names (e.g., *Proc.* for Proceedings).
- Include all publication details, such as volume, issue, page range, and year.

Reference List Format

1. Books:

[1] Author(s) Initial(s). Surname(s), *Title of Book*, xth ed. City of Publisher, (only U.S. State), Country: Publisher, year.

Example:

[1] J. P. Hailman, *Coding and Redundancy: Man-Made and Animal-Evolved Signals*. Cambridge, MA, USA: Harvard Univ. Press, 2008.

2. Journal Articles:

[2] Author(s) Initial(s). Surname(s), "Title of Paper," Abbrev. *Title of Journal*, vol. x, no. x, pp. xxx-xxx, Month, year.

Example:

[2] J. Kou, "Estimating the number of clusters via the GUD statistic," *J. Comput. Graph. Statist.*, vol. 23, no. 2, pp. 403–417, Jun. 2014, doi: 10.1080/10618600.2013.778778

3. Conference Papers:

[3] Author(s) Initial(s). Surname(s), "Title of Paper," in Abbrev. *Title of Conf.*, City of Conf., (only U.S. State), Country, year, pp. xxx-xxx.

Example:

[3] C. D. Author and E. F. Author, "Title of paper," in Proc. IEEE *Int. Symp. Circuits Syst.*, New Orleans, LA, USA, 1990, pp. 123-126.

4. Online Sources:

[4] Author(s) Initial(s). Surname(s), "Title of Web Page," Website Name. [Online]. Available: <http://www.website.com>. [Accessed: Month Day, year].

Example:

[4] B. Fung. "Amazon offers concessions to resolve EU antitrust probes." CNN.com. <https://edition.cnn.com/2022/07/14/tech/amazon-concessions-eu-antitrust/index.html> (accessed Jul. 18, 2022).

APPENDICES

- Include additional data, calculations, and supplementary material relevant to the thesis.